KANAK MAHESHWARI

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PROFESSIONAL SUMMARY

Master of Information Technology graduate from **UNSW Sydney** specializing in **Artificial Intelligence**, with industry experience developing **Backend Systems**, **RESTful APIs**, and **AI-powered Applications**. Experienced in building **distributed backend services**, **asynchronous processing pipelines**, implementing **Retrieval-Augmented Generation (RAG)** solutions, and integrating **Large Language Models (LLMs)** into production-oriented applications. Skilled in **Python, TypeScript, React, Django, FastAPI, Docker, Git, Cursor,** and **Claude Code**, with experience delivering scalable software products across technology and consulting environments.

TECHNICAL SKILLS

Languages:	Python, TypeScript, SQL, C++, Java
Backend:	FastAPI, Django, Flask, REST APIs, Celery, Redis
AI & LLMs:	LangChain, LangGraph, FAISS, Ollama Prompt Engineering, Groq
Frontend:	React, Tailwind CSS
Databases:	SQLite, PostgreSQL, MongoDB
Tools & DevOps:	Docker, Azure, AWS, Git, Render, Vercel, Cursor, Claude Code, Codex, GitHub Copilot
Big Data:	Apache Spark, PySpark, distributed data processing

EDUCATION

University of New South Wales (UNSW Sydney) — **Sydney, Australia**
Master of Information Technology - AI — *Aug 2026*

VIT Bhopal University — **India**
Bachelor of Technology in Computer Science and Engineering - AI and Machine Learning — *May 2024*

PROFESSIONAL EXPERIENCE

UNSW — **Sydney**
Tutor — *Sept 2025 - Present*

- Conducted tutorials covering **algorithms, data structures, and C programming**, guiding undergraduate and postgraduate students through core computer science concepts.
- Evaluated assignments and provided constructive feedback on **code optimization**, **time and space complexity analysis**, and programming best practices.
- Mentor students in **debugging**, **software design principles**, and systematic **problem-solving** techniques through interactive tutorials and one-on-one support.

Octane Software Solutions — **Sydney**
Intern — *Dec 2025 - Feb 2026*

- Developed AI-powered models and automation workflows to support practical business processes and improve operational efficiency.
- Applied Python and IBM AI technologies to develop and test intelligent solutions, gaining hands-on experience with enterprise AI and automation tools.
- Collaborated within a technical team to develop, test, and refine AI-driven solutions while strengthening debugging, problem-solving, and software development skills.

PwC — **Mumbai, India**
Intern — *Feb 2024 - Apr 2024*

- Developed a machine learning solution to analyze employee datasets and generate data-driven employee benefit recommendations.
- Built data preprocessing and feature engineering pipelines using Python, Pandas, SQL, and Excel to prepare enterprise datasets for model training.
- Evaluated model performance through data analysis and visualization, supporting informed business decision-making.
- Worked collaboratively with consulting teams to deliver AI-driven insights for employee benefit optimization.

SELECTED PROJECTS

WorkTrace AI - AI-Powered Workflow Intelligence Platform

- Engineered an **Electron desktop application** to capture screen activity, user input events, and optional microphone audio for reconstructing real-world workflows.
- Built a **FastAPI backend** with authentication, tenant isolation, chunk-upload validation, workflow session management, and privacy sanitisation.
- Developed an asynchronous **AI processing pipeline** to analyse captured workflow evidence, generate pointer annotations, align workflow steps, and produce structured **SOPs**.
- Integrated **Whisper-based audio transcription** to combine spoken context with visual workflow evidence during workflow reconstruction.
- Built workflow review interfaces for **processing status, transcripts, workflow metrics, generated SOPs, and annotated workflow steps**.

NLtoRegex

- Designed and developed a distributed backend platform for large-scale data transformation using **Django** and **RESTful APIs**.
- Engineered asynchronous processing pipelines with **Celery** and **Redis** to efficiently handle long-running file processing tasks.
- Integrated **Large Language Models (LLMs)** to translate natural language instructions into executable **regular-expression workflows**.
- Developed scalable **REST APIs** for file uploads, job scheduling, execution tracking, and result retrieval.
- Containerized backend services using **Docker** to enable reproducible development and deployment across distributed environments.

Today's News Application

- Developed a high-performance **FastAPI** backend for intelligent news aggregation, processing, and retrieval.
- Built an automated pipeline to extract, clean, categorize, summarize, and store news articles from multiple sources.
- Designed scalable **FastAPIs** enabling semantic search, AI-powered summarization, and conversational interactions over news content.
- Integrated **SQLite** for structured data storage and implemented **RAG** using **FAISS** for efficient vector similarity search, enabling scalable retrieval of news content.

QurioPilot - AI Powered Learning Platform

- Engineered **FastAPI** backend services powering AI-driven personalized learning workflows.
- Integrated locally hosted **LLaMA** models through **Ollama** to deliver real-time conversational AI capabilities.
- Designed scalable **FastAPIs** connecting the React frontend with AI inference services and personalized learning modules.
- Developed interactive learning features, including AI-generated quizzes, adaptive learning paths, and educational mini-games to enhance user engagement.
- Deployed backend services on **Render** and frontend applications on **Vercel** for end-to-end application delivery.

ACHIEVEMENTS & LEADERSHIP

- Achieved **Second Place** at a UNSW CSESoc Hackathon for developing QurioPilot, an AI-powered learning platform.
- Served as football team captain for two consecutive years and led the team to first place.
- Competed as a state-level swimmer.